INCH-POUND
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DETAIL SPECIFICATION

FLOWMETERS, MECHANICAL, VOLUMETRIC POSITIVE DISPLACEMENT, LIQUID, COLD WATER AND HOT WATER TYPE, FOR NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fresh water (see 6.5.2) mechanical flowmeters (liquid, cold and hot water service) of the volumetric positive displacement type for naval shipboard use.

1.2 Classification. Flowmeters are of the following classes, as specified (see 6.2):

a. Class A – Volumetric flow measurement of cold fresh water up to 100 °F (38 °C).

b. Class B – Volumetric flow measurement of hot fresh water from 100 to 180 °F (38 to 82 °C).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads – UN and UNR Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-889 - Dissimilar Metals

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.24 - Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves: Classes 150, 300, 600, 900, 1500, and 2500

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

(Copies of this document are available online at www.astm.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and 6.3.

3.2 Description. The flowmeter shall consist of a volumetric-type measuring drum, or a fixed displacement device, and a mechanically or magnetically actuated registering element. The assembly comprising the flowmeter shall be equipped for installation in a shipboard piping system. The liquid shall be measured by isolating the flow either in a predetermined volume space or in a fixed displacement mechanism, such as a piston assembly or nutating disk assembly. Flowmeter parts in contact with the measured liquid shall be composed of materials that will not chemically react with the liquid. Unless otherwise specified (see 6.2), the measured liquid shall be fresh or potable water (see 6.5.3). Unless otherwise specified (see 6.2), the flowmeter shall not be supplied with a resettable counter. The flowmeter parts shall be arranged to operate within the accuracy requirements specified herein and shall be readily accessible for maintenance and replacement without disturbing the connections to the pipe lines.

3.3 Approved standard commercial product. The flowmeter shall be, as a minimum, the manufacturer's approved standard commercial product (see 6.5.1) and shall meet the requirements of this specification. Additional or improved features not specifically prohibited by this specification, but which are a part of the manufacturer's approved standard commercial product may be included.

3.4 Material. Material shall be as specified herein. Material not specified shall be of the quality normally used in the manufacturer's approved standard practice. Material shall not have been in prior service except as required by manufacturer's tests.

3.4.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, and other means in accordance with MIL-STD-889. Bolts, nuts, pins, screws, and other fasteners shall be of the same material as the metals joined or shall be cathodic to the metals joined.

3.4.2 Prohibited materials. Flowmeters shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs. Lead and its compounds shall not be used where they can come in contact with fresh water. The lead content in bronze alloys shall not exceed levels allowed by ASTM B61.

3.4.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Construction. The flowmeters shall be constructed to maintain accuracy of indication and to resist wear and corrosion. Materials for construction of the pressure-containing parts shall be selected to be compatible with the pressure and temperature ratings of this specification. Moving parts shall be properly aligned, balanced, and adequately lubricated. Interior parts, seals, and packings shall be accessible for service without disturbing the connections to the pipe lines.

3.6 Components. The flowmeter shall consist of additional apparatus including the components specified in 3.6.1 through 3.6.8 and those normally furnished with the commercial types of these flowmeters.

3.6.1 Flowmeter case. The flowmeter case shall be constructed of a non-ferrous, non-magnetic, and non-corrosive alloy.

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3.6.2 Register case. The register case and lid shall be constructed of a non-ferrous, non-magnetic, and non-corrosive alloy. In addition, it shall be removable but securely fastened to the flowmeter case so as not to present a missile hazard under MIL-DTL-901 shock conditions specified in 4.6.4 and to ensure maximum protection to the register. The register window shall be fabricated from a clear, non-fogging material of sufficient size to allow for visualization of the entire register face.

3.6.3 Measuring chamber. The measuring chamber shall be designed to maintain a constant volume displacement or constant volumetric measurement with each cycle of the meter. Unless otherwise specified (see 6.2), it shall be constructed of the manufacturer's approved standard material.

3.6.4 Piston or disk measuring elements. Piston or disk-type measuring elements shall be supported and aligned to prevent twisting, bending, and distortion. Piston nutations or oscillations shall be kept to a minimum, consistent with standard industry practice.

3.6.5 Register. Unless otherwise specified (see 6.2), the register shall be of the horizontal type, with figures reading in United States liquid gallons (USG) direct, or with both USG and liters. The respective maximum indication unit of the initial register circle or wheel, and the respective minimum indication of the register shall be as specified in table I. The flowmeter designations specified in table I are representative of ranges of volumetric flow rates for class A and class B flowmeters as specified in 3.7.1.2 and 3.7.2.2, respectively. The flowmeter designations do not necessarily correspond to a manufacturer's flowmeter designation which is specified in terms of line, pipe, or model size. The correlation between flowmeter designation and nominal pipe size is specified in 3.6.8.

TABLE I. Flowmeter register indication.

Flowmeter designation	Maximum indication unit of initial register circle or wheel (gallons)	Minimum indication of register (gallons)
1	10	999,990
2	10	999,990
3	10	999,990
4	100	999,990
5	100	9,999,900
6	100	9,999,900
NOTE: 1 gallon equals 3.8 liters.		

3.6.6 Strainer. Unless otherwise specified (see 6.2), the flowmeter shall be furnished with integral strainers. Strainer screens shall be rigid, fit snugly, be easy to remove, and have an effective straining area at least double that of the main case inlet.

3.6.7 Check valve. When a reversal of flow through the flowmeter might result in damage, the flowmeter shall be furnished with a check valve or other means for preventing such flow.

3.6.8 Piping connections. Unless otherwise specified (see 6.2), class A and class B flowmeters shall have Nominal Pipe Size (NPS) end connections. Screw threads may be used up through NPS 2 and shall be National Pipe Thread (NPT) in accordance with FED-STD-H28 and FED-STD-H28/2. Flanged connections shall be machined on faces and rim with bolt holes spotfaced and drilled in accordance with ASME B16.24 for 150-pound copper alloy (bronze) flanges.

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3.7 Performance characteristics.

3.7.1 Class A flowmeters.

3.7.1.1 Working pressure. Unless otherwise specified (see 6.2), class A flowmeters shall be capable of a minimum working pressure of 150 pounds per square inch (lb/in^2) (10.3 bar) (see 4.6.1).

3.7.1.2 Registration. The registration (see 6.5.4) of class A flowmeters shall indicate not less than 98 nor more than 102 percent of the water actually passed through the flowmeter within the limits of the minimum and maximum flow rates specified in table II. Minimum flow rates shall be not more than 10 percent of the maximum flow rate. There shall be not less than 90 percent registration of the water which actually passes through the flowmeter at a flow rate of 5 percent of the maximum flow rate (see 4.6.2). These registration limits shall be maintained at all water temperatures up to 100 °F (38 °C).

3.7.1.3 Pressure drop. The pressure drop through any class A flowmeter shall not exceed 25 lb/in^2 (1.7 bar) at the maximum flow rates specified in table II (see 4.6.3).

TABLE II. Class A flowmeter capacities.

Class A flowmeter designation	Maximum flow rate (gallons/minute)	Continuous flow rate (gallons/minute)	Minimum flow rate (gallons/minute)
1	10	5	1
2	30	15	3
3	70	35	7
4	150	75	15
5	500	250	50
6	1000	500	100
NOTE: 1 gallon/minute equals 3.8 liters/minute.			

3.7.2 Class B flowmeters.

3.7.2.1 Working pressure. Unless otherwise specified (see 6.2), class B flowmeters shall be capable of a minimum working pressure of 100 lb/in^2 (6.9 bar) (see 4.6.1).

3.7.2.2 Registration. The registration of class B flowmeters shall indicate not less than 97 nor more than 103 percent of the water actually passed through the flowmeter within the limits of minimum and maximum flow rates specified in table III. Minimum flow rates shall be not more than 20 percent of the maximum flow rate. There shall be not less than 90 percent registration of the water which actually passes through the flowmeter at a flow rate of 10 percent the maximum flow rate. These registration limits shall be maintained at all water temperatures between 100 and 180 °F (38 to 82 °C) (see 4.6.2.2).

3.7.2.3 Pressure drop. The pressure drop through any class B flowmeter shall not exceed 20 lb/in^2 (1.4 bar) at the maximum flow rates specified in table III (see 4.6.3).

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TABLE III. Class B flowmeter capacities.

Class B flowmeter designation	Maximum flow rate (gallons/minute)	Continuous flow rate (gallons/minute)	Minimum flow rate (gallons/minute)
1	10	5	2
2	20	10	4
3	50	25	10
4	100	50	20
5	250	125	50
6	500	250	100
NOTE: 1 gallon/minute equals 3.8 liters/minute.			

3.7.3 Class A and B flowmeters.

3.7.3.1 Flow rates. Class A and B flowmeters shall have a maximum flow rate of not less than the maximum flow rate specified in tables II or III, as applicable. The minimum flow rates shall not be more than the minimum flow rates specified in tables II or III, as applicable.

3.7.3.2 Continuous flow rate. Class A and B flowmeters shall have a continuous flow rate of not less than 50 percent of the maximum flow rates specified in tables II or III, as applicable.

3.7.3.3 Intermittent flow. Intermittent flow for class A and B flowmeters shall be considered as operation at the maximum flow rate for not longer than 2 hours out of every 8-hour period.

3.7.4 Shock. The flowmeter shall be resistant to mechanical shock and shall not present any secondary missile hazard (pieces flying or coming off) when tested in accordance with the grade B, class II, lightweight shock requirements of MIL-DTL-901 as specified in 4.6.4.

3.8 Frost protection. When specified (see 6.2), flowmeters shall be furnished with a frost protection device, such as a deformable air bag or replaceable rupture plate, designed to yield under normal freezing conditions to protect flowmeter parts. Providing external insulation is not considered a frost protection device.

3.9 Marking.

3.9.1 Identification markings. Flowmeters shall bear the manufacturer's serial number permanently marked on the meter body and on the register box lid. Flowmeters shall bear the manufacturer's name, manufacturer's model designation, normal flow range, and the designation "Class A" or "Class B", as applicable. Meter markings shall be in accordance with MIL-STD-130.

3.9.2 Direction-of-flow indication. Flowmeters shall have the direction of flow indicated in raised characters on the flowmeter body.

3.10 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.11 Reliability and maintainability. Each flowmeter system shall have an operating life of 100,000 hours with no scheduled maintenance. Corrective maintenance shall be no more than 1 hour per 10,000 hours of operation. Special tools required for maintaining the flowmeter shall be as specified (see 6.2). Special tools are defined as tools not listed in the GSA Global Supply Catalog. Replacement or repair parts and service for the approved standard commercial product provided under this specification shall be available from the manufacturer through commercial parts distribution or service organizations. Technical manuals shall be as specified (see 6.2).

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3.12 Workmanship. The quality of workmanship shall be such as to produce flowmeters that are in accordance with the requirements of this specification and that are constructed to ensure proper functioning of all parts of the flowmeter.

3.12.1 Castings and forgings. Castings and forgings shall have no defects or scale and shall not be mismatched. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste, shall not be attempted.

3.12.2 Treatment and painting. Unless otherwise specified (see 6.2), the flowmeter shall be treated and painted in accordance with the manufacturer's standard practice.

3.12.3 Surface finish. Surfaces of castings, forgings, or molded parts, stampings, and welded parts shall be clean and free of sand, dirt, scale, flux, or other extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one flowmeter when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4, 4.5, and the tests of 4.6 as specified in table IV. The first article may be a standard production item from the contractor's current inventory, provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnished under the contract.

4.3 Conformance inspection. Conformance inspection shall be performed on sample flowmeters selected in accordance with 4.3.2. This inspection shall include the examination of 4.4 as specified in table IV.

4.3.1 Inspection lot. Flowmeters of the same class and size, offered to the Government at one time, shall be considered a lot for purposes of inspection.

4.3.2 Sampling for examination. A random sample of flowmeters shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.3 Sampling for tests. Sampling is not applicable. Flowmeters passing the examination of 4.4 shall be tested in accordance with 4.6.

TABLE IV. First article and conformance inspections.

Characteristic	Requirement	Verification	First article inspection	Conformance inspection
Description – general requirements	3.1	4.4	X	X
Approved standard commercial product	3.2	4.4	X	X
Material	3.4	4.4	X	
Prohibited materials	3.4.2	4.5	X	
Construction	3.5	4.4	X	
Components	3.6	4.4	X	
Working pressure	3.7.1.1 3.7.2.1	4.6.1	X	

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TABLE IV. First article and conformance inspections – Continued.

Characteristic	Requirement	Verification	First article inspection	Conformance inspection
Registration	3.7.1.2, 3.7.2.2	4.6.2.2	X	
Pressure drop	3.7.1.3, 3.7.2.3	4.6.3	X	
Flow rates	3.7.3.1	4.6.2.2	X	
Continuous flow rate	3.7.3.2	4.6.2.2	X	
Intermittent flow	3.7.3.3	4.6.2.2	X	
Shock	3.7.4	4.6.4	X	
Frost protection	3.8	4.4	X	X
Markings	3.9	4.4	X	X
Interchangeability	3.10	4.6.5	X	
Reliability and maintainability	3.11	4.6.5	X	
Workmanship	3.12	4.4	X	X

4.4 Examination. Flowmeters selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification and those of their manufacturer. Examination shall be conducted as specified in table V. Defective flowmeters shall be rejected and, if the number of defective flowmeters in any sample exceeds the acceptance number for that sample, the lot represented shall be rejected.

TABLE V. Classification of defects.

Categories	Defects
Critical	None defined.
--	--
Major	
101	Class not as specified.
102	Missing, broken, or improper parts.
103	Flowmeters not rated in accordance with system requirements.
104	Materials not as specified.
105	End connections not as specified.
106	Maximum indication unit of initial register circle or wheel and respective minimum capacity or register nonconforming.
107	Horizontal type registers not furnished with covers; do not protect delicate and fragile parts.
108	Dial face difficult to read, cloudy, fogs easily, not adequately protected.
109	Loose, unbalanced, or improperly meshing gears.
110	Defective castings or forgings.
111	Marking not as specified.
113	Wearing parts not accessible for service.
114	Strainers furnished do not fit; size of strainer opening incorrect.

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TABLE V. Classification of defects – Continued.

Categories	Defects
115	Check valve to prevent flowmeter damage during flow reversal not furnished or not functioning properly.
116	Frost protection (if specified, see 3.8 and 6.2) not present.
--	--
Minor	
201	Surface finish not as specified.
202	Painted surfaces scratched or uncovered.

4.5 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4.2 and 6.2).

4.6 Methods of inspection testing.

4.6.1 Hydrostatic leakage. Sample flowmeters selected in accordance with 4.3.3 shall be subjected to an internal hydrostatic pressure for 30 minutes equal to 150 percent of the working pressure specified in either 3.7.1.1 or 3.7.2.1, as applicable.

4.6.2 Calibration. Sample flowmeters selected in accordance with 4.3.3 shall be subjected to a calibration as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 Test conditions. Class A flowmeters shall be tested at a water temperature of 95±5 °F (35±3 °C) at 5, 10, 15, 25, 50, 75, and 100 percent of the maximum flow rate specified in table II. Class B flowmeters shall be tested at water temperatures of 105±5 °F (40±3 °C) and 175±5 °F (79±3 °C) at 10, 15, 20, 25, 50, 75, and 100 percent of the maximum flow rate specified in table III.

4.6.2.2 Registration. Registration shall be verified in accordance with 3.7.1.2 or 3.7.2.2, as applicable, at the test conditions specified in 4.6.2.1. Satisfactory tests conducted at these conditions shall verify the requirements of 3.7.3.1, 3.7.3.2, and 3.7.3.3.

4.6.3 Pressure drop. Sample flowmeters selected in accordance with 4.3.3 shall be tested to verify conformance to 3.7.1.3 or 3.7.2.3, as applicable.

4.6.4 Shock. The flowmeter shall be subjected to grade B, class II, lightweight classification, type B shock in accordance with MIL-DTL-901.

4.6.5 Reliability and maintainability. The flowmeter shall be disassembled, and all components shall be reassembled and replaced. The personnel and specific types of tools used in the demonstration shall be identified. In addition, the demonstration shall verify that the requirements of 3.11 are satisfied by the design and construction of the flowmeter. The demonstration shall be conducted utilizing the technical manual and the recommended tools without the use of an expert mechanic. The demonstration procedures in MIL-HDBK-470 may be used. Particular attention shall be directed toward the possibility of loss of small parts by the maintenance crew. The validation procedures in MIL-HDBK-217 may be used. The test shall be considered a failure if the flowmeter does not meet the maintainability, reliability, and repair parts requirements of 3.11, when examined as specified in 4.4 or if the flowmeter is unable to be disassembled and reassembled without personnel hazard in accordance with procedures outlined in the technical manual and restored to full operation.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The flowmeters covered by this specification are intended for operation in fresh water systems at temperatures up to 180 °F (82 °C), for flow rates up to 1000 gallons per minute (3800 liters per minute), and meet Navy shipboard shock requirements. Current stock of flowmeters may be used until depleted; future requisitions for flowmeters should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size of flowmeter (see 1.2 and tables II and III).

c. When first article inspection is required (see 3.1).

d. Liquid to be used if not fresh or potable water (see 3.1).

e. When a resettable counter is required (see 3.2).

f. Requirement for prohibited materials verification (see 3.4.2 and 4.5).

g. When the measuring chamber is not to be constructed of the manufacturer's approved standard material (see 3.6.3).

h. Type of register, if other than specified (see 3.6.5).

i. Type of strainer, if other than specified (see 3.6.6).

j. Type of end connections, if other than specified (see 3.6.8).

k. Working pressure, if other than specified (see 3.7.1.1 or 3.7.2.1).

l. When a frost protection device is required (see 3.8).

m. Special tools required for maintaining the flowmeter (see 3.11).

n. When technical manuals are required (see 3.11).

o. Treatment and painting requirements, if other than specified (see 3.12.2).

p. AQL of 1.0 (see 4.3.2).

q. Packaging requirements (see 5.1).

r. Requirements for Provisioning Technical Documentation (PTD), spare parts, and repair parts (see 6.4).

6.3 First article inspection. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be as specified (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Definitions.

6.5.1 Approved standard commercial product. The manufacturer's approved standard commercial product is a product that has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures and represents the latest approved production models. The principal contracting officer will determine compliance with the provision of part II of the Federal Acquisition Regulation.

6.5.2 Fresh water. Fresh water is any naturally occurring water except seawater and brackish water, generally characterized by having low concentrations of dissolved salts and other total dissolved solids.

6.5.3 Potable water. Potable water is fresh water that has been made safe for human consumption.

6.5.4 Registration. Registration is the ratio of the volume of water indicated on the register to the volume of water that actually passes through the flowmeter at a specified flow rate.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
(Project 6680-2020-001)

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